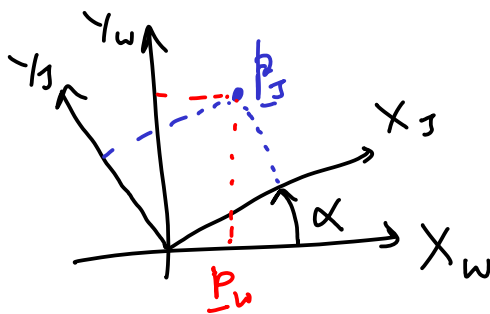


Rotation matrix



Jetbot w.r.t.
World is α

$$\underline{p}_w = {}^w R_J \underline{p}_J$$

$$\downarrow$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_J \\ y_J \end{bmatrix}$$

① Orthonormal matrix
 ${}^w R_J(\alpha)$

$${}^w R_J^T {}^w R_J = \underline{I}$$

$${}^w R_J^{-1} = {}^w R_J^T = {}^J R_w$$

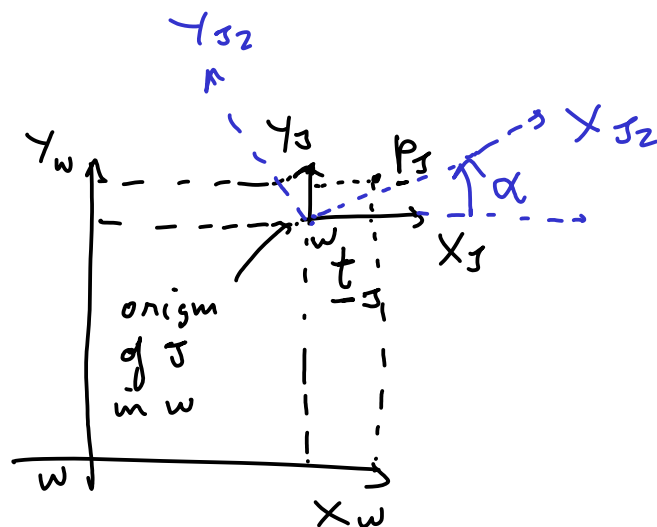
$${}^w R_J^{-1} = {}^w R_J(-\alpha)$$

$$\textcircled{2} \det({}^w R_J) = 1$$

Rotation + translation

$$\underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w \xleftarrow{{}^w \underline{t}_J} \underline{p}_J \xleftarrow{R(\alpha)} \underline{p}_{J_2}$$



$$\textcircled{1} \underline{p}_J = {}^J R_J(\alpha) \underline{p}_{J_2}$$

$$\textcircled{2} \underline{p}_w = \underline{p}_J + {}^w \underline{t}_J$$

$$\underline{p}_w = {}^J R_J(\alpha) \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

↑ Affine transformation

Transformation matrix

$$\underline{p}_w = {}^J R_{J_2} \underline{p}_{J_2} + {}^w \underline{t}_{J_2}$$

$$\begin{bmatrix} x_w \\ y_w \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \cos \alpha & -\sin \alpha & t_x \\ \sin \alpha & \cos \alpha & t_y \\ 0 & 0 & 1 \end{bmatrix}}_{{}^w T_{J_2}} \begin{bmatrix} x_{J_2} \\ y_{J_2} \\ 1 \end{bmatrix}$$

homogeneous coordinates

${}^w T_{J_2}$

$$\Rightarrow \boxed{\underline{\tilde{p}}_w = {}^w T_{J_2} \underline{\tilde{p}}_{J_2}} \Leftrightarrow \begin{bmatrix} \underline{p}_w \\ 1 \end{bmatrix} = \begin{bmatrix} {}^w R_{J_2} & {}^w \underline{t}_{J_2} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{p}_{J_2} \\ 1 \end{bmatrix}$$

Block-wise matrix multiplication works as long as the ^{component} multiplications are valid

Properties of Transformation matrix

① Find the inverse of a transformation matrix using the fact that $R^{-1} = R^T$

$$T = \begin{bmatrix} R_{2 \times 2} & t_{2 \times 1} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}_{3 \times 3}$$

$$T^{-1} = ?$$

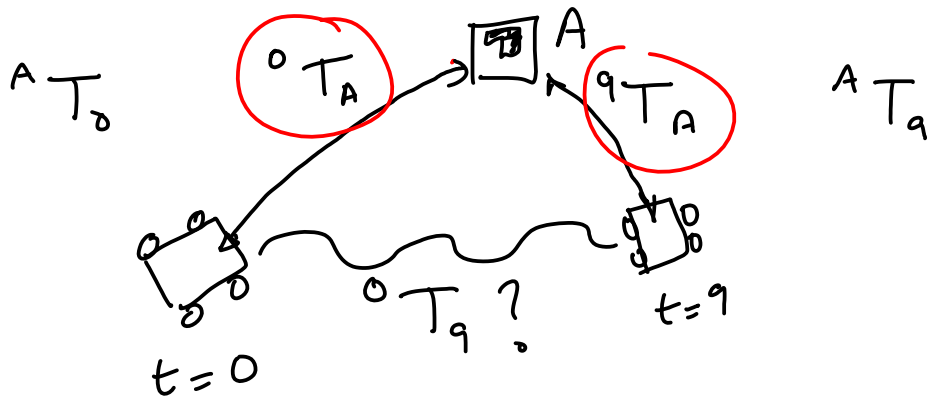
$$\begin{aligned} \underline{y} &= R \underline{x} + \underline{t} \\ \underline{x} &= R^T (\underline{y} - \underline{t}) \end{aligned}$$

$$\begin{aligned} \underline{y} &= T \underline{x} \\ \underline{x} &= T^{-1} \underline{y} \end{aligned}$$

$$\underline{x} = R^T \underline{y} - [R^T \underline{t}]_{2 \times 1}$$

$$\begin{bmatrix} \underline{x} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R_{2 \times 2}^T & -R_{2 \times 1}^T \underline{t} \\ \underline{0}_{1 \times 2}^T & 1 \end{bmatrix}}_{T^{-1}} \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix}$$

$$T = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \Rightarrow T^{-1} = \begin{bmatrix} R^T & -R^T \underline{t} \\ \underline{0}^T & 1 \end{bmatrix}$$



$${}^0T_q = ?$$

$$\underline{x}_0 = {}^0T_q \underline{x}_q$$

$$\underline{x}_A = {}^AT_q \underline{x}_q$$

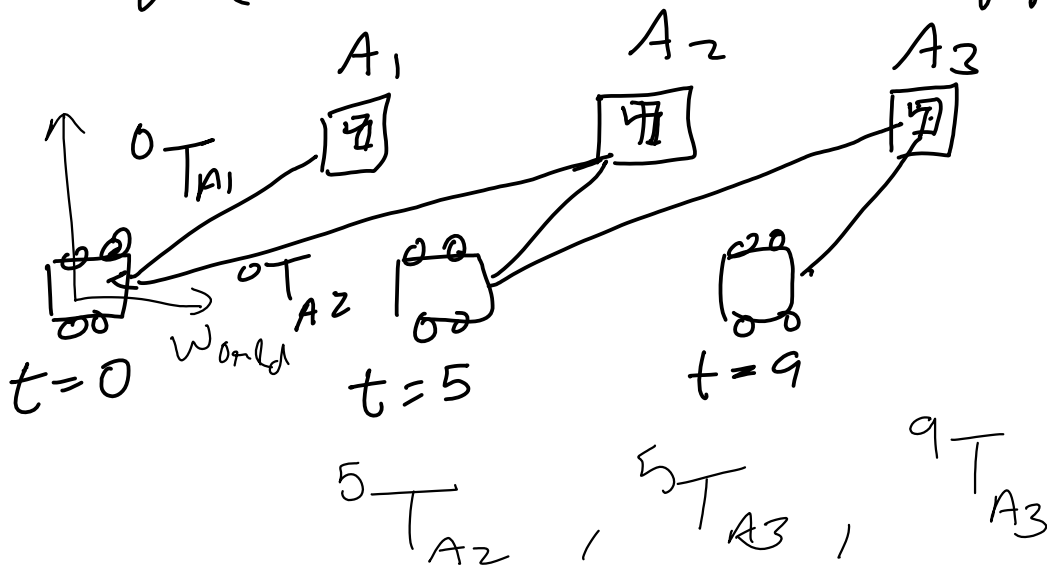
$$\underline{x}_0 = {}^0T_A \underline{x}_A$$

$$\underline{x}_0 = \underbrace{{}^0T_A {}^AT_q}_{{}^0T_q} \underline{x}_q$$

$$\Rightarrow \boxed{{}^0T_q = {}^0T_A {}^AT_q}$$

Composition of transformations

Mapping (Landmark based mapping)



$${}^0T_{A3} = ?$$

$${}^0T_q = ?$$

